CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 15: Similarity (With Applications to Maps and Models)

Every formula and result of the chapter on one sheet

1	Similar triangles	Equal corresponding angles; proportional corresponding sides
2	AA condition	Two pairs of angles equal
3	SAS condition	One pair of angles equal and the including sides proportional
4	SSS condition	Three pairs of sides proportional
5	Basic proportionality theorem	If $DE \parallel BC$: $\frac{AD}{DB} = \frac{AE}{EC}$
6	Parallel line and the third side	$\frac{DE}{BC} = \frac{AD}{AB} = \frac{AE}{AC}$
7	Ratio of areas	$\frac{\text{area}_1}{\text{area}_2} = \left(\frac{\text{side}_1}{\text{side}_2}\right)^2$
8	Ratio of perimeters, altitudes, medians	Equal to the ratio of corresponding sides
9	Scale factor	$k = \frac{\text{length on the map or model}}{\text{actual length}}$
10	Area with scale factor k	Area of model $= k^2 \times$ actual area
11	Volume with scale factor k	Volume of model $= k^3 \times$ actual volume
12	Right triangle with altitude to the hypotenuse	$AB^2 = AD \times AC$ and $BD^2 = AD \times DC$